

Computer Programming (NCSV101) [Winter 2025-26]

Extra Slides# 2

Virtual Machines

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Operating Systems are life-savers

They save you from talking to the hardware directly

They perform a lot of tasks that you would otherwise have to do on your own

But “how” they do it differs from one OS to another

What they let you do with your hardware also differs from one OS to another

Linux, for example, is “usually” considered closer to the hardware

- ... meaning that you may have more finer-grained control on your hardware

Windows and Macintosh, are known to be closer to the user

- ... meaning that they may “hide” finer-grained details of how the hardware is managed

OS on top of OS – Virtual Machines

What if I like one OS a lot, but I am stuck with another OS?

- For example, you *paid* for a copy of Windows with your laptop... and you just can't let the money go waste :P

There are some software solutions which can help you

They are called *Hypervisors*

A Hypervisor is a collection of programs which can *emulate* hardware !!

Basically, they make a *layer* over your OS

- This layer behaves pretty much the same way that the hardware does

So, you can now *install* an OS over this *emulated* hardware

- We call this emulated hardware a **Virtual Machine** – because it is not real, but virtual
- ... and this OS can be different from the OS running directly over your hardware

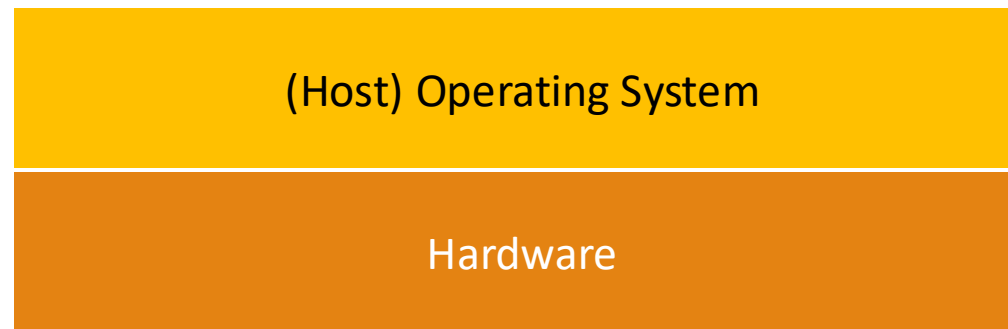
OS on top of OS – Virtual Machines



Hardware

This is your hardware

OS on top of OS – Virtual Machines

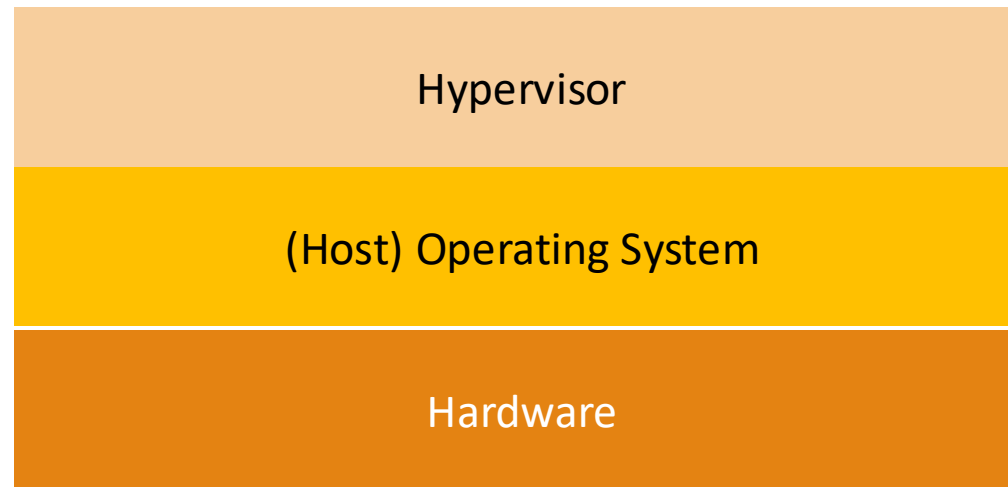


You install an Operating System over it

OS on top of OS – Virtual Machines

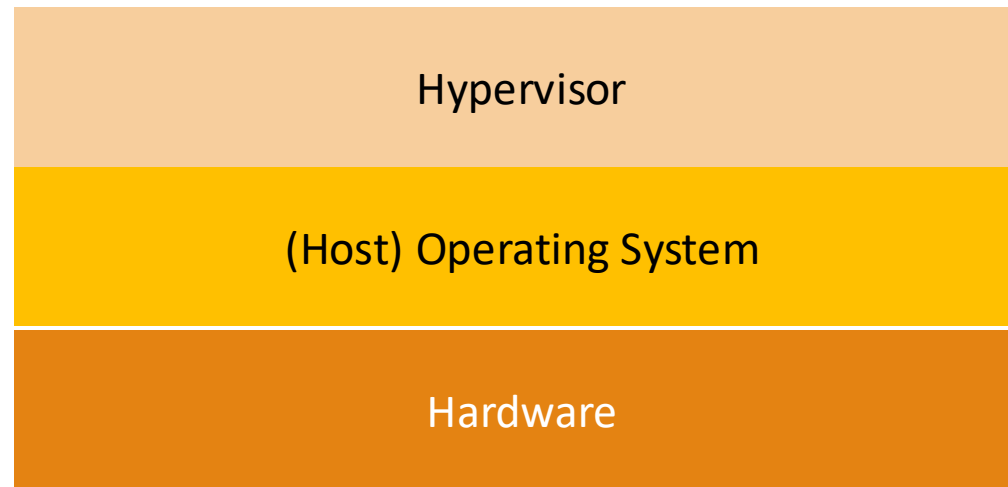


OS on top of OS – Virtual Machines



You install a Hypervisor over your Operating System

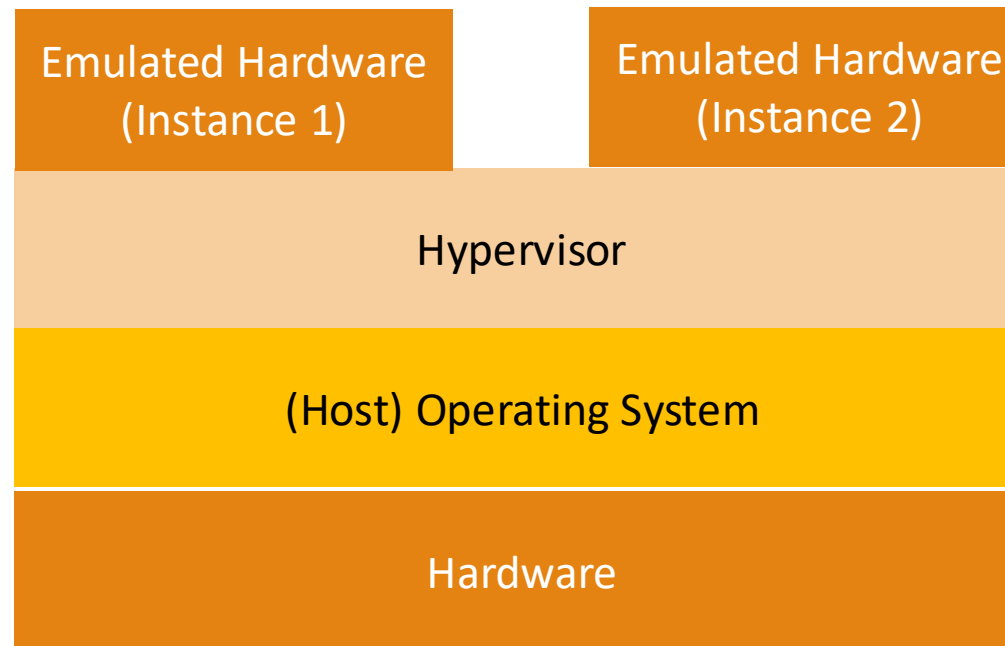
OS on top of OS – Virtual Machines



Just like you install any other software tool

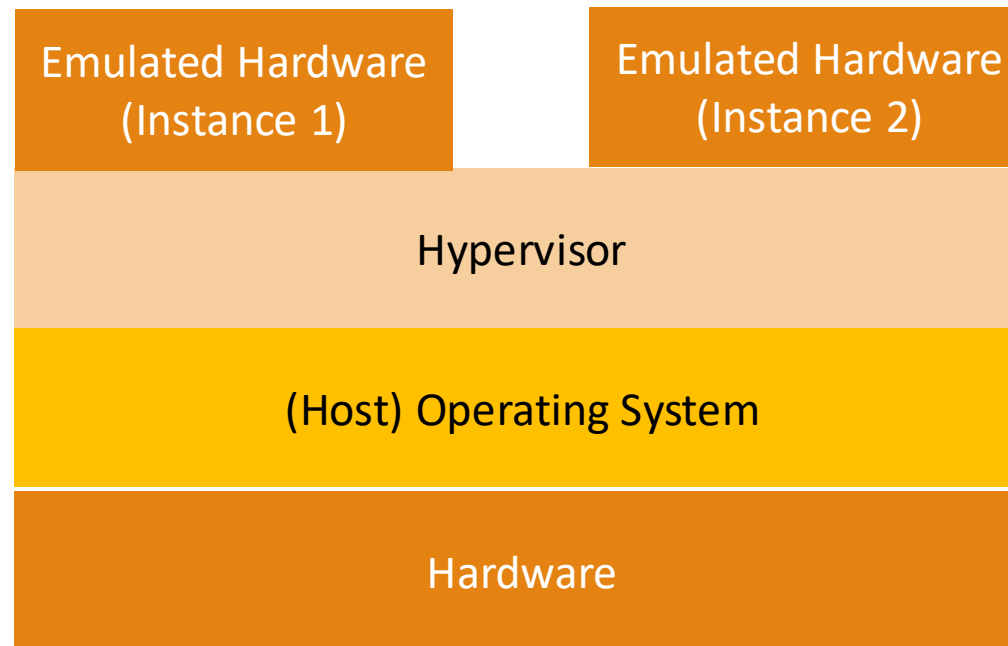
You install a Hypervisor over your Operating System

OS on top of OS – Virtual Machines



The Hypervisor can create one or more hardware emulations

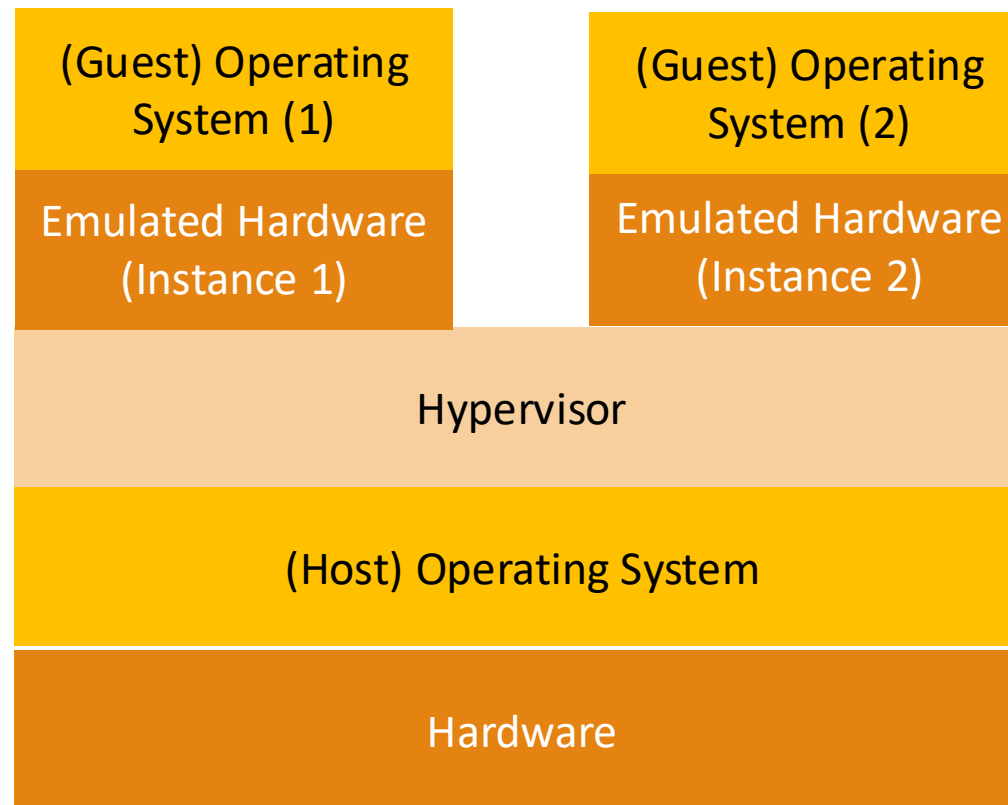
OS on top of OS – Virtual Machines



Each emulation emulates every type of essential hardware – CPU, I/O, Main Memory etc.

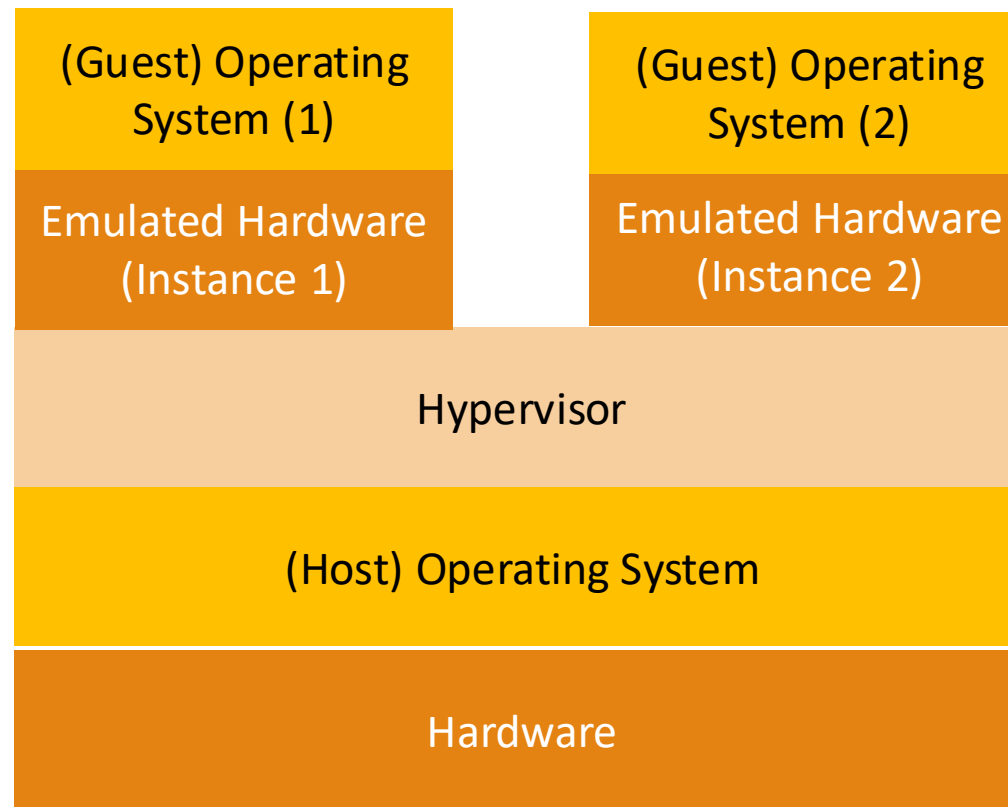
The Hypervisor can create one or more hardware emulations

OS on top of OS – Virtual Machines



You can now install one or more Operating Systems over the emulated hardware

OS on top of OS – Virtual Machines



We call these Operating Systems *guest* Operating Systems

You can now install one or more Operating Systems over the emulated hardware

Using Virtual Machines for the Course

If for some reason, you cannot use a Linux flavour as your host OS, use a Virtual Machine

Install a Hypervisor on your OS

- I use Oracle Virtualbox
- Another option is VMware Workstation Player

Create a Virtual Machine over it

Additional Reading !!

What we saw today is known as *type-2* Hypervisor

- There is another type of Hypervisor, called *type-1* Hypervisor

Read about the difference between the two

- Reading this short article should be enough:
<https://www.ibm.com/cloud/learn/hypervisors>

There is one awesome feature of Virtual Machines – Snapshots

- Read more about snapshoting
- This article provides a basic introduction:
<https://www.techrepublic.com/article/how-to-use-snapshots-in-virtualbox/>